



Prediction of the proximate analysis parameters of refuse-derived fuel based on deep learning approach

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Abstract

Determination of proximate characteristics can be achieved using conventional analyses methods that require a certain amount of time. In cement factories, refuse-derived fuel (RDF) is continuously fed to a kiln by a conveyor belt, so even if an inappropriate proximate characteristic is determined, it would be too late to prevent the feeding of RDF to the kiln. To overcome this problem, there is a need for instant measurement of the proximate characteristics (moisture, volatile matter, ash) that enables the feeding to be stopped. In such cases, the deep learning (DL) is a useful method based on the prediction of proximate characteristics. Therefore, in this study, the aim is to estimate the mentioned parameters developed by near-infrared spectroscopy (NIR) combined with deep learning models. For this purpose, the spectrographic measurements taken from RDF samples with an NIR spectrometer, and the results of proximate analysis in a laboratory, were used together as a dataset. A fully convolutional neural network (FCNN) and ResNet were used as a network, and they were trained using images of RDF samples and proximate analysis values. The FCNN model was more successful in prediction studies. According to the FCNN model, the results show that the models in the study can predict the moisture, ash, and volatile matter content of RDF with satisfactory R^2 values between 0.979, 0.983, and 0.952.

Keywords Ash content · Deep learning · Moisture content · Refuse-derived fuel (RDF) · Volatile matter content

Introduction

Cement production is an energy-intensive process that needs a remarkable amount of coal consumption (Hekal et al. 2021). In addition to coal consumption, it also plays an important role in the generation of an industrial based greenhouse impact. The substitution of coal with alternative fuels will take place in the strategies of cement producers,

based on the Paris Agreement by 2030, and European Green Deal Climate Neutrality by 2050 objectives. According to the data shared by the European Cement Association (CEMBUREAU), a 46% alternative fuel use in cement kilns avoids 18 million tonnes of CO₂ eq. emissions annually (Cembureau 2021). Refused-derived fuels (RDF) prepared from municipal/industrial waste are alternative energy materials consumed by the cement industry to substitute for coal. This processing includes the removal of non-flammable materials (such as glass, metal, and stone) to improve the burning behavior of the waste and to produce a product with a more uniform particle size distribution. The combustion behavior of RDF depends on its moisture, ash, and volatile matter content, also known as the proximate characteristic, as it is for coal. According to the Turkish Communique on refuse-derived fuels, additional fuels, and alternative raw material (29,036/20.06.2014), the moisture content of RDF should be lower than 35%, while the calorific value should be higher than 2500 kcal/kg. In order for the heating value to be at these values, the ash content must be low, and the volatile matter content must be high.

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Table 1 Prediction studies of moisture, ash, and volatile matter content in fuels based on the DL and ML methods

Reference	Fuel type	Predicted component	DL method combination
Xiao and Le 2020	Coal	Moisture, ash, volatile matter	Visible infrared spectroscopy (VIS)-deep belief net
Qin et al. 2019	Coal	Volatile matter content	Laser-induced breakdown spectroscopy (LIBS)–Fourier transform infrared spectroscopy (FT-IR) and partial least-squares (PLS) (ML) model
Le et al. 2018	Coal	Moisture, ash, volatile matter	VIS-CNN
Gómez et al. 2018	Coal	Ash, volatile matter	FT-IR photoacoustic spectroscopy (PAS)–PLS (ML)
Hu et al. 2015	Coal	Ash, volatile matter	NIR spectroscopy with LS-SVM and PLSR (ML)
Andres and Bona 2005	Coal	Moisture, ash, volatile matter	Diffuse reflectance infrared Fourier transform spectroscopy (DRIFT)–NIR spectra–PLS (ML)
Lu et al. 2021	Wood pellets	Ash, volatile matter	LIBS–PLS (ML)
Sgarbossa et al. 2014	Wood pellets	Moisture, ash	Classification of the of wood pellet samples depending on the color (RGB and HSV) tone
Posom and Sirisomboon 2017	Ground bamboo	Ash, volatile matter	Fourier transform NIR spectroscopy–PLS (ML)
Rambo et al. 2020	Coconut and coffee residues	Moisture, ash, volatile matter	NIR spectroscopy–PLS (ML)
Posom and Sirisomboon 2015	Jatropha curcas kernels	Moisture	Fourier transform NIR spectroscopy–PLS (ML)
Posom et al. 2016	Leucaena leucocephala pellets	Moisture	NIR spectroscopy–PLS (ML)

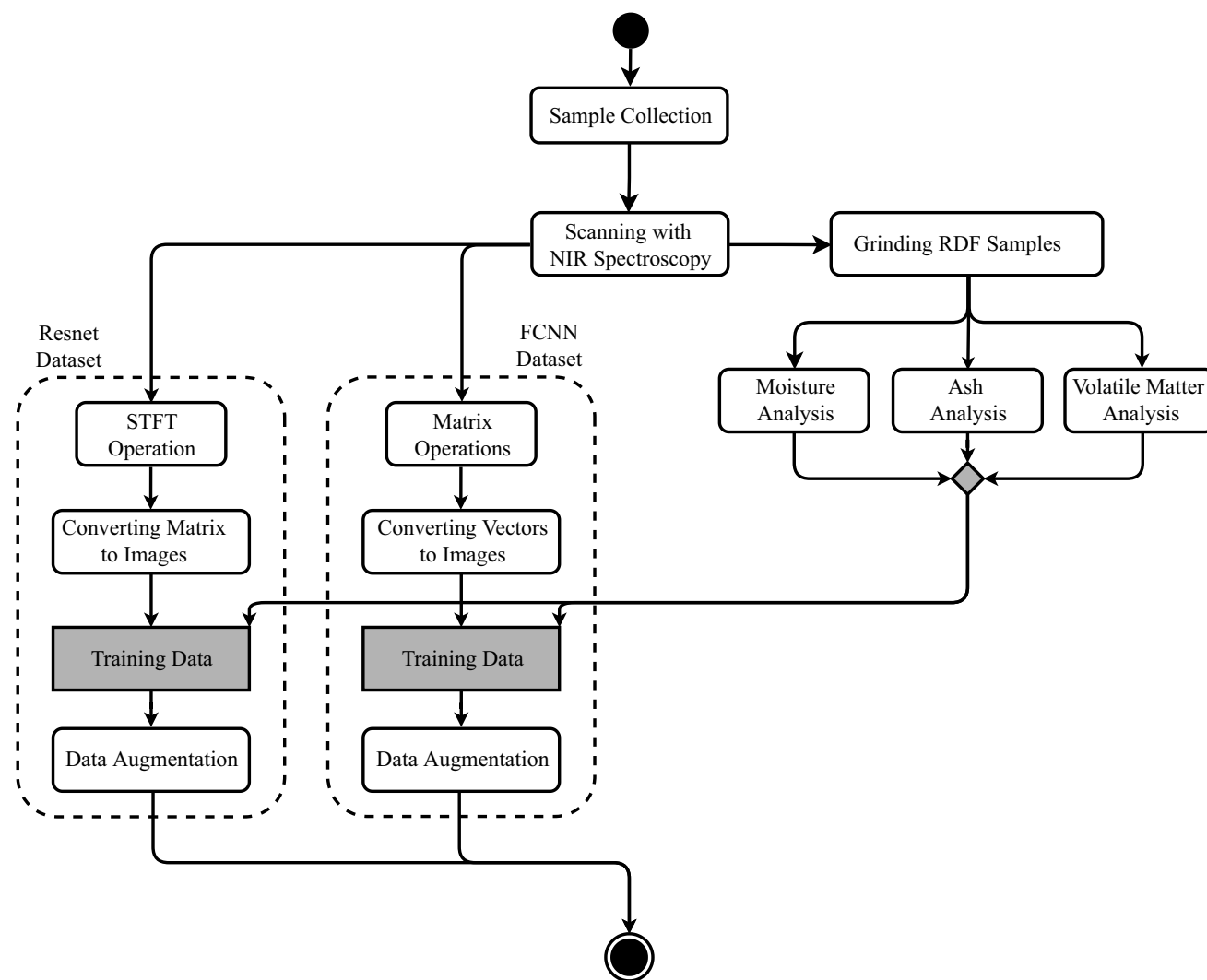
**Fig. 1** Flow chart of the study



Fig. 2 RDF sample

Deep learning (DL) is a computational technique that utilizes multiple hidden processing layers for learning data representations and relationships with multiple levels of abstraction. Traditional machine learning (ML) algorithms have limitations when it comes to processing unprocessed natural data. A technique known as “representation learning” enables a computer to be fed with unstructured data and

automatically find the representations required for learning. Deep learning (DL) is a representation learning technique with many layers of representation that is generated by building simple, but non-linear modules, that each changes the representation at one level (beginning with the raw input) into a higher level. DL provides excellent performance above the ability of the human eye and produces accurate results such as the advanced image processing methods (Toğaçar et al. 2020). Table 1 refers to the prediction studies on fuels by using DL and machine learning (ML) methods.

As can be seen from Table 1, many prediction studies for proximate analysis are available for fuels and biomass, but no study has been found for RDF. Starting from this point, this study aims to close this gap by presenting a combination of hyperspectral imaging and DL methods (fully convolutional neural network, FCNN, and ResNet) to predict the moisture, ash, and volatile matter content of RDF.

Methodology

The study was carried out following the flow chart given in Fig. 1. Each part is explained in detail in the following sections.

Sample collection

RDF samples were collected from an RDF processing plant next to a cement factory in Eskişehir, Turkey. The RDF samples consist of industrial waste with a high calorific content, such as non-recyclable packaging waste, contaminated paper, plastics, textile products, and paint waste. In order to

Fig. 3 A sample for spectrometric measurements

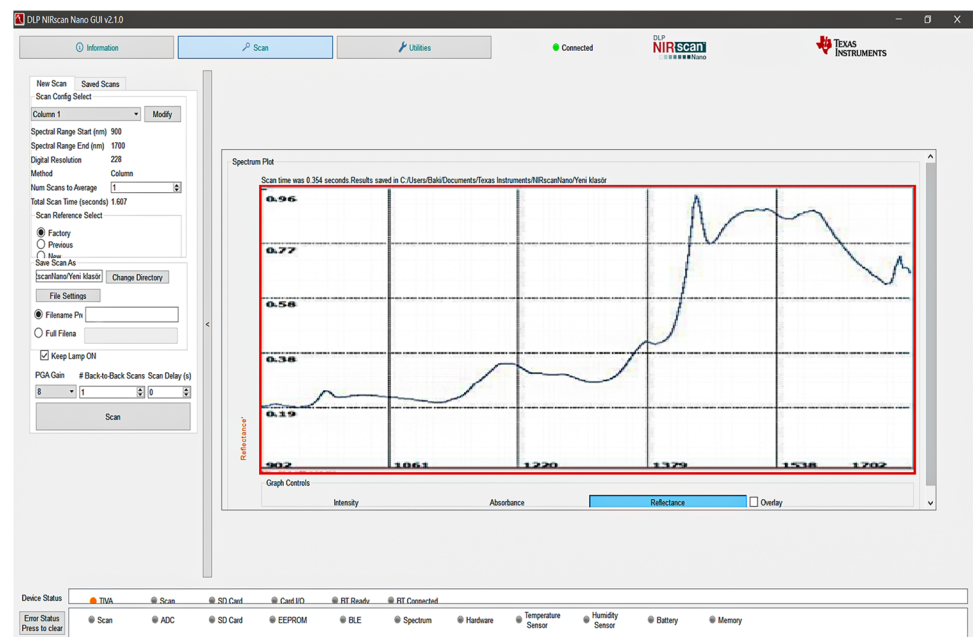
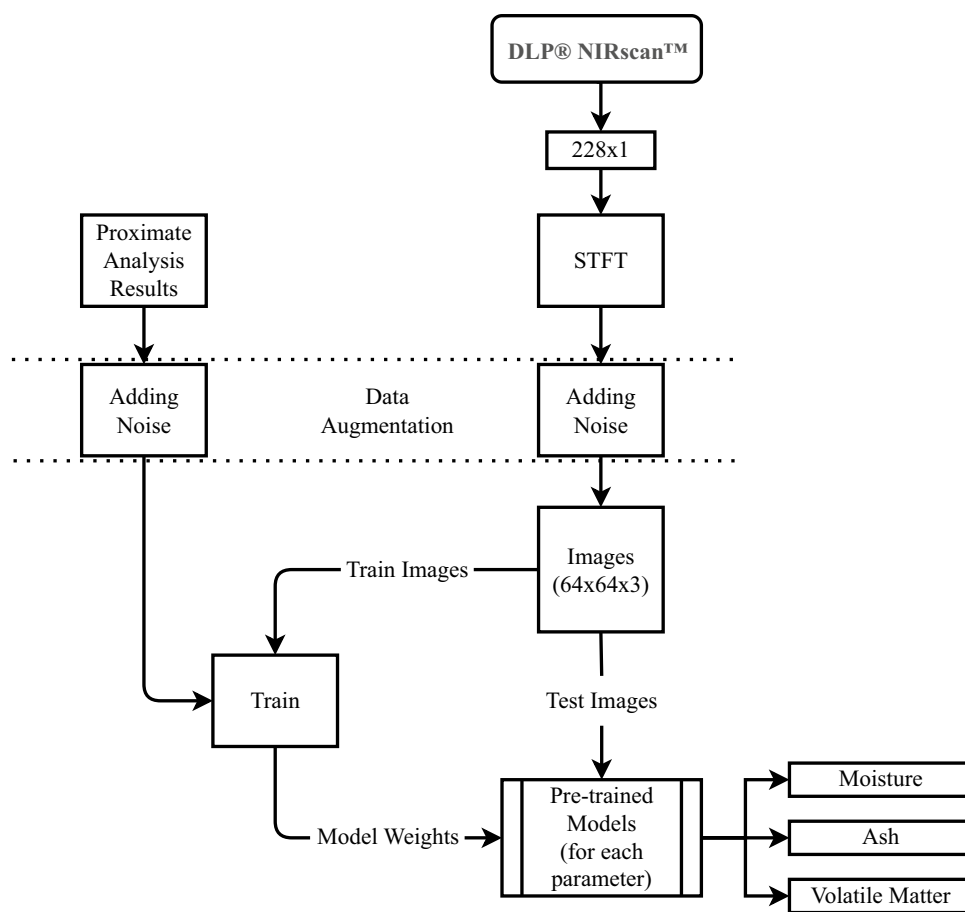


Fig. 4 STFT and the ResNet model operating diagram



provide diversity in image processing, 55 RDF samples, in the range of 100–150 g with heterogeneous content, were chosen from different points of the stockpile in the RDF plant. Samples were taken in sample bags at different times and were used for moisture, ash, and volatile matter analysis (Fig. 2).

NIR reflectance spectra scanning

The spectrometer reflectance values of the samples were used for modeling. For this purpose, measurements were taken with a DLPNIRNANOEVN NIR spectrometer (TEXAS Instruments TIDA 00,554) (available at <https://www.ti.com/tool/DLPNIRNANOEVN>) from certain points on the RDF by shifting the spectrometer. Each measurement data included 228 reflectance values taken at equal intervals (approximately 3.5 nm) from a specific light wavelength range (900–1700 nm) supported by the NIR spectrometer. The data set was augmented by adding noise to the image and label values. Measurements were repeated five times for each of the collected samples and used separately. The entered data was compensated by subtracting it from the mean in order not to be affected by the changes between the

measurements. In this way, the background values in the measurements are eliminated. The screenshot of the spectrometer user interface for one of the measurements is given in Fig. 3.

Proximate (moisture, ash, and volatile matter) analysis

After the NIR spectra of the obtained RDF samples were taken with an NIR spectrometer, the moisture, ash, and volatile matter analyses of the 55 samples were measured according to ASTM E 790–15 2016; ASTM D3174-12 2018; ASTM-D3175-20 2020, respectively. The statistical analysis of the experimental results, and measurement uncertainties for each parameter, were calculated based on three replicates.

Statistical evaluation

The statistical evaluation of the results was achieved by drawing frequency histogram graphs for each parameter based on the three repetitions for 55 of the RDF samples. The mean, mode, median, standard deviation, coefficient of

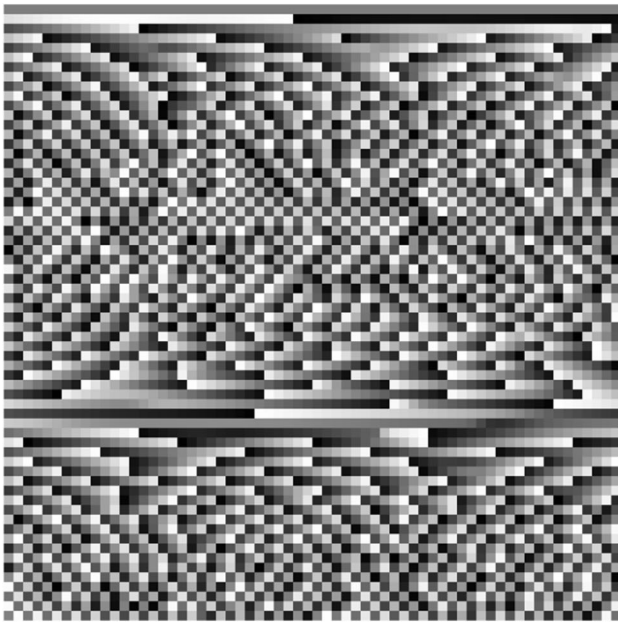


Fig. 5 The ResNet train set image sample

variation, coefficient of skewness, and coefficient of kurtosis values were calculated using frequency data and Eqs. 1–7; where, $\bar{x}$ (mean value), f_i (frequency), x_i (the midpoint of the i th data range), and n (number of observations); Med (median); s (standard deviation).

$$\text{Mean} : \bar{x} = \frac{\sum f_i x_i}{n} \quad (1)$$

$$\text{Mode} : 3(\text{Med}) - 2(\bar{x}) \quad (2)$$

$$\text{Standard deviation} : s = \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{n-1}} \quad (3)$$

$$\text{Coefficient of variation} : CV = \frac{100s}{\bar{x}} \quad (4)$$

$$\text{Coefficient of skewness} : \alpha_3 = \frac{\sum f_i (x_i - \bar{x})^3 / n - 1}{s^3} \quad (5)$$

$$\alpha'_3 = \frac{(\bar{x} - \text{mode})}{s} \quad (6)$$

$$\text{Coefficient of kurtosis} : \alpha_4 = \frac{\sum f_i (x_i - \bar{x})^4 / n - 1}{s^4} \quad (7)$$

The mean is defined as the arithmetic mean of a series of individual or calculated evaluations. If a series of observations is arranged in order of increasing value, the middle most observation in the series or the arithmetic mean of the middle two observations in the series is known as the median. The value that occurs at the highest frequency in a series of observations is known as the mode. Although the standard deviation can be used as an indicator of the absolute distribution of a range of measured values, it provides little information about whether the value is large or small. For this reason, the calculation of the coefficient of variation is of great importance (Tchobanoglous et al. 1993).

Measurement uncertainty

The measurement uncertainty affects the efficiency of the prediction. For this reason, uncertainty that would be generated from the measurements were presented as expanded measurement uncertainties, since for most purposes in analytical chemistry, an expanded uncertainty, U , should be used. The expanded uncertainty provides an interval within which the value of the measure is believed to lie with a higher level of confidence. U is obtained by multiplying the combined standard uncertainty by a coverage factor, k . For an approximate level of confidence of 95%, k is usually set to 2. In estimating the combined uncertainty, it is necessary to take each parameter taking place in the calculation equation of the result as a source of uncertainty. The moisture, ash, and volatile matter content calculations are based on gravimetric analyses, and the calculation equations are based on the first and second weighing. The expanded uncertainty

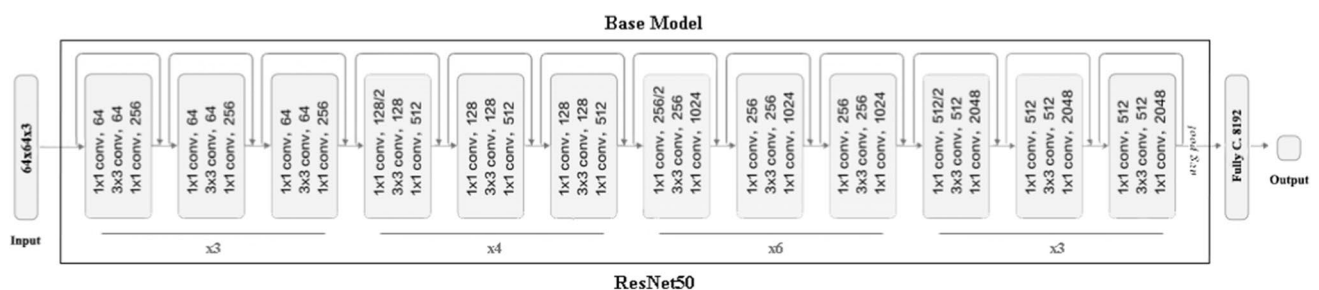


Fig. 6 General structure of the model

calculation based on the weighing was defined as the calibration uncertainty using the uncertainty of the analytical balance (0.1 mg, 95%, $k=2$). The other, which may be the most important uncertainty component, is the repeatability.

The expanded uncertainty resulted from repeatability was calculated using the standard deviation and relative standard deviation (that is, standard deviation as a percentage of the mean value) (EUROCHEM/CITAC 2012). The minimum

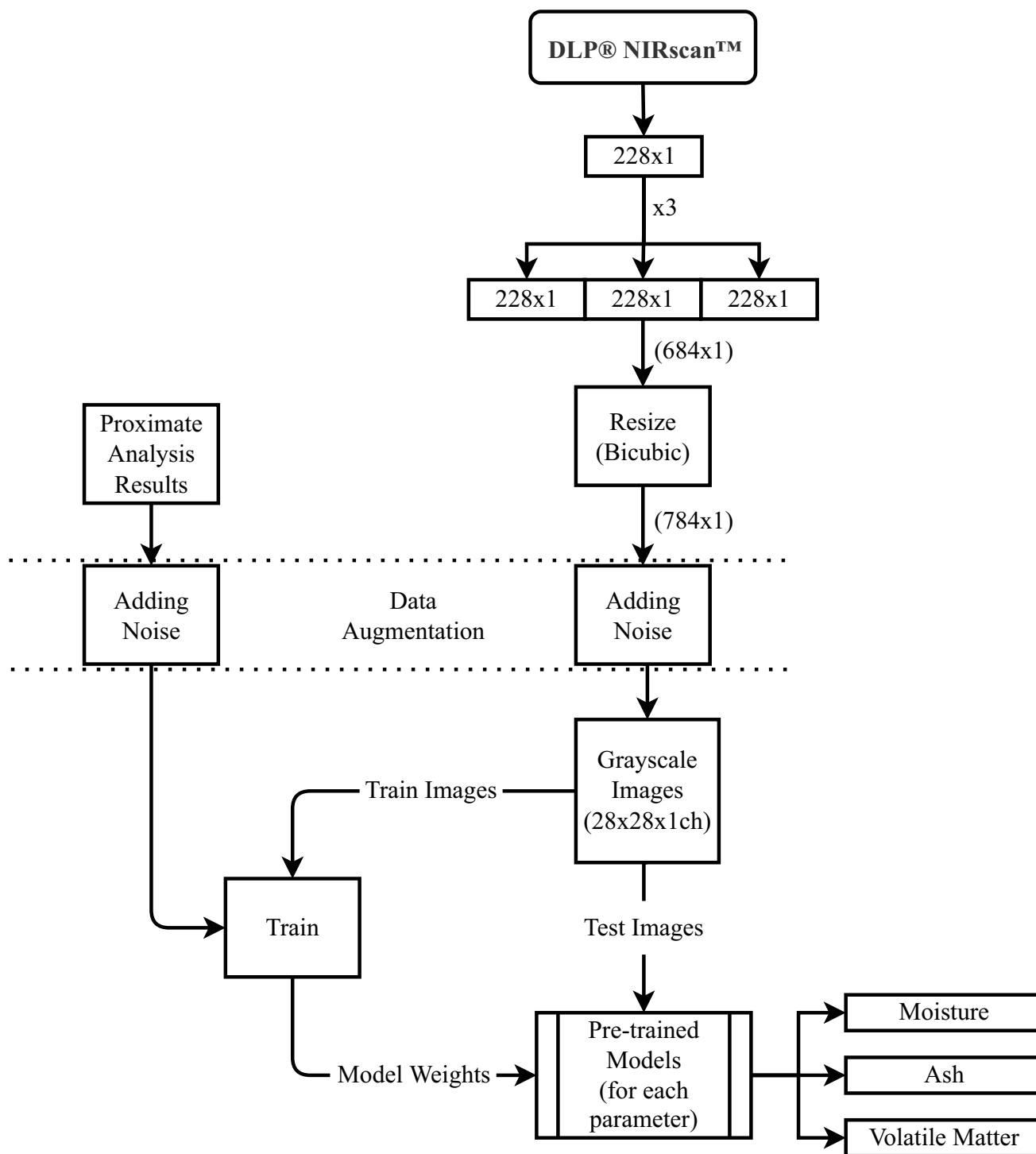


Fig. 7 FCNN model operating diagram

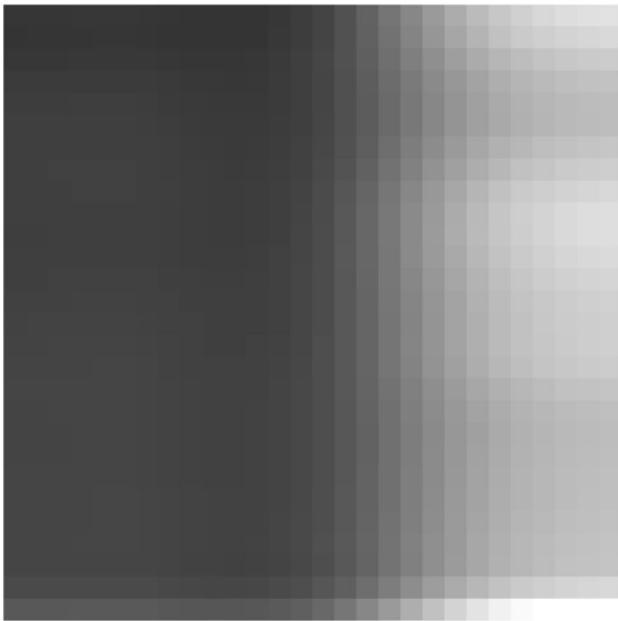


Fig. 8 The FCNN train set image sample

and maximum values that samples can take according to the measurement uncertainty results were considered in modeling studies with deep learning.

Deep learning modeling methods and development of NIR data

In this study, two different deep learning modeling methods are used; ResNet with short time Fourier transform (STFT) and the FCNN model. The fact that the data collected in the NIR spectroscopy is in wave form led to the use of STFT for extracting more features in the deep learning phase. STFT is a Fourier transform used to determine the sinusoidal

frequency and phase content of local parts of wave signals. ResNet was preferred in the training of these data obtained by STFT. The deep neural network, developed by the Microsoft Research Team, with the changes in the convolutional neural network (CNN), architecture achieved important advantages, such as training the deep network faster and reaching a network structure with less parameters because of its orientation to a deep network structure (He et al. 2016). Additionally, in the study, spectrometer data in waveform are transformed into visuals with matrix operations and modeled separately. At this stage, fully convolutional neural network (FCNN) was chosen, because it is successful in visual recognition and can be used in value estimation with the fully connected final layer.

Coding of the models was conducted on Tensorflow using the Keras library. The computer configuration of the experiment platform is i7-6700HQ CPU, 16 GB ram, 64 bits Ubuntu 18.04, and Nvidia Gtx 960 m GPU. Detailed explanation of the models is given in two sub-titles.

ResNet with STFT

In this part of the study, the data set was created with STFT and the ResNet model was applied. The operation diagram of the system is given in Fig. 4.

Firstly, the data set was created by applying STFT to the measurement data taken separately for each sample in the NIR imaging section. STFT is a Fourier transform used to determine the sinusoidal frequency and phase content of local parts of a time varying signal. STFTs are obtained by dividing a long signal into shorter segments of equal length and then calculating the Fourier transform of each short segment separately. The formula of STFT is given in Eq. 8.

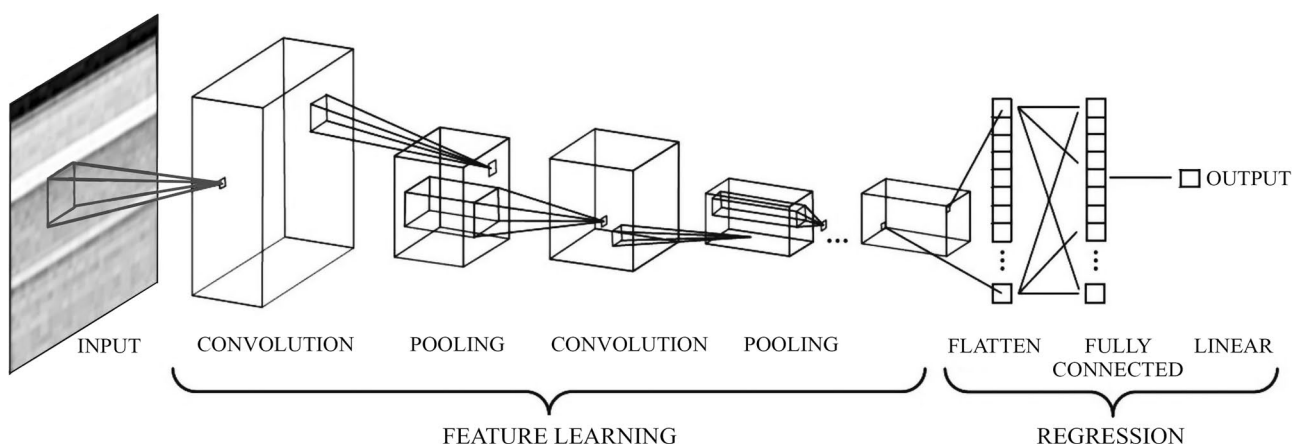


Fig. 9 General structure of the FCNN model

Table 2 Proximate analysis data in the literature and this study

Reference	RDF content	Proximate analysis (%)		
		Moisture	Ash	Volatile matter
Miskolczi et al. 2011	Plastics, papers, and other waste (rice, stems, wood, tin foil, and other waste from households)	4.1	15.5	80.4
Efika et al. 2015	Plastic	4.2	12.8	73
Zhao et al. 2016	Plastic, paper/cardboard, textiles, horticulture and food waste, landfill mining materials, and chicken manure	73.8	1.1	21.4
Akdağ et al. 2016	Textile, paper, and plastics	1.6	12.9	81.8
Liedmann et al. 2017	Paper, plastic, textile product, foil, and fine particles	13.53	9.09	71.54
Nakhaei et al. 2018	Plastic, wood, paper, cardboard, textile, and fine particles (RDF A)	11.3	13.1	66.0
Nakhaei et al. 2018	Contains less plastic, wood, paper, and cardboard than RDF A (RDF B)	19.8	15.5	45.9
Nobre et al. 2019	Non-hazardous industrial waste	10.5	16.9	74.9
Hashem et al. 2019	Rubber	4.8	19.2	80.8
El-Salamony et al. 2020	RDF combined with rice husks at different percentages (paper and cardboard 12%, plastic 29%, tissues 13%, garden waste 40%, wood and its products 6%; RDF A)	20.2	20.4	73.5
El-Salamony et al. 2020	RDF combined with rice husks at different percentages (plastic 31%, tissues 24%, garden waste 43%, wood and its products 2%; RDF B)	15.2	19.0	79.4
El-Salamony et al. 2020	RDF combined with rice husks at different percentages (paper and cardboard 8%, plastic 42%, tissues 17%, garden waste 10%, wood and its products 9%, rocks and stones 14%; RDF C)	25.2	27.6	59.3
El-Salamony et al. 2020	RDF combined with rice husks at different percentages (paper and cardboard 8%, plastic 43%, tissues 26%, garden waste 9%, wood and its products 9%, rocks and stones 5%) (RDF D)	21.3	25.6	67.3
El-Salamony et al. 2020	RDF combined with rice husks at different percentages (paper and cardboard 12%, plastic 20%, tissues 8%, wood and its products 60%) (RDF E)	23.5	25.9	64.2
This study		0.04–4.28	2.84–18.22	63.37–89.36

$$\text{STFT}_x(\tau, f) = \int_{-\infty}^{+\infty} x(t)h(t - \tau)e^{-j2\pi ft} dt \quad (8)$$

This multiplication transforms the signal into a two-dimensional time–frequency plane, where $x(t)$ is the time-based signal field and $h(t - \tau)e^{-j2\pi ft}$ is the basic function of STFT. The *scipy.signal.stft* function in the *Scipy* library, given in Eq. 9, was used while STFT was applied to the data (SciPy 2021).

$$f, t, Z = \text{signal.stft}(\text{HspectData}, fs = 1, nperseg = 128, noverlap = 125, nfft = \text{None}) \quad (9)$$

f array of sample frequencies

t array of segment times

Z STFT of x

HspectData hyperspectral data

fs sampling frequency of the x time series

$nperseg$ length of each segment

$noverlap$ number of points to overlap between segments

$nfft$ the length of the FFT used: if none, the FFT length is $nperseg$

The Z matrix of 65×67 dimensions acquired as a result of STFT was trimmed and reduced to 64×64 dimensions. Since the model we will use uses 3-channel images, this data has been copied to all channels. An example of the created images is given in Fig. 5. The laboratory analysis results were added to the data set to use as labels during the training.

After the data creation part was completed, it was seen that the data obtained were not sufficient for training. Deep learning requires a lot of data during the training phase, but analysis of the samples in the laboratory is not easy and takes time. In such cases, data augmentation methods are used. The data set has been augmented by adding noise to the image and label values.

The modeling part is based on ResNet50, but the input and output layers have been changed to fit the data. Since the standard ResNet model works with $224 \times 224 \times 3$ images, the

Layer (type)	Output Shape
conv2d_1 (Conv2D)	(None, 28, 28, 8)
batch_normalization_1 (Batch Normalization)	(None, 28, 28, 8)
activation_1 (Activation)	(None, 28, 28, 8)
average_pooling2d_1 (Average Pooling)	(None, 14, 14, 8)
conv2d_2 (Conv2D)	(None, 14, 14, 16)
batch_normalization_2 (Batch Normalization)	(None, 14, 14, 16)
activation_2 (Activation)	(None, 14, 14, 16)
average_pooling2d_2 (Average Pooling)	(None, 7, 7, 16)
conv2d_3 (Conv2D)	(None, 7, 7, 32)
batch_normalization_3 (Batch Normalization)	(None, 7, 7, 32)
activation_3 (Activation)	(None, 7, 7, 32)
conv2d_4 (Conv2D)	(None, 7, 7, 32)
batch_normalization_4 (Batch Normalization)	(None, 7, 7, 32)
activation_4 (Activation)	(None, 7, 7, 32)
dropout_1 (Dropout)	(None, 7, 7, 32)
flatten_1 (Flatten)	(None, 1568)
dense_1 (Dense)	(None, 1)

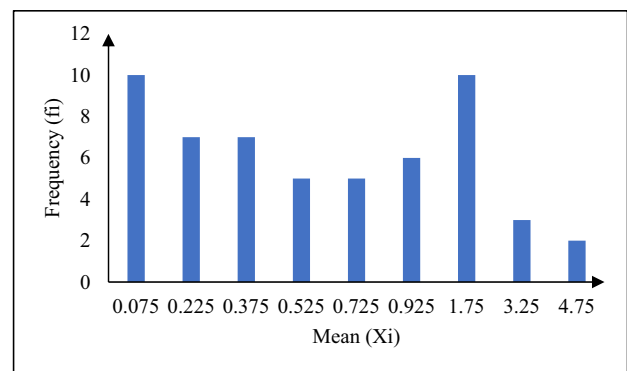
Fig. 10 Layers of the FCNN model

input layer is adapted for our $64 \times 64 \times 3$ data. In addition, since the output layer will estimate the value instead of classification, the output layer is arranged with a linear activation function. The output layer is used to predict the amount of moisture, ash, and volatile matter in the RDF samples. The general structure of the model is given in Fig. 6.

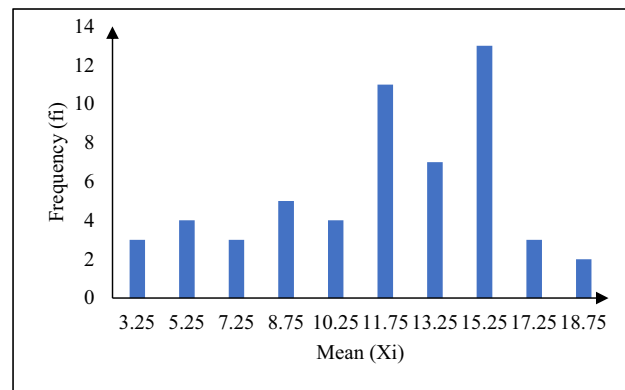
FCNN model

In this part of the study, different than the previous one, NIR imaging data was converted directly into visuals and remodeled using FCNN. The operation diagram of the system is given in Fig. 7.

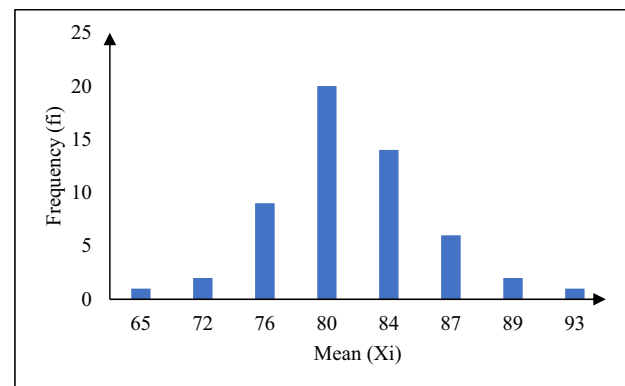
Firstly, for use with FCNN, vectors of size 684×1 were obtained by adding triple permutations of five measurements sized 228×1 one after the other. In order to convert these obtained matrices into single channel images measuring 28×28 , bicubic interpolation was carried out into 784×1 matrices. During the image conversion process, normalization in the range of 0–255 and C-contiguous order reshape processes were applied. An example of the created images with numerical differences is given in Fig. 8. The



(a)



(b)



(c)

Fig. 11 Histogram graphs of the results of the proximate analysis. (a) Moisture values. (b) Ash values. (c) Volatile matter values

laboratory analysis results were added to the data set to use as labels during the training. The data set was augmented by adding noise to the image and label values.

In this part of the study, a model using CNN with a fully connected last layer was proposed. The general structure of the FCNN model is given in Fig. 9. The layers of the model are detailed in Fig. 10. The convolution operation produces different numbers of feature maps, depending on the numbers of filters used. The pooling operations reduce the spatial

Table 3 Statistical analysis results of the proximate analysis data

Parameter	Moisture, %	Ash, %	Volatile matter, %
Mean	0.97	11.76	81.13
Mode	3.30	23.73	77.75
Median	1.75	15.25	80
Standard deviation	0.091	0.31	0.43
Coefficient of variation	9.33	2.64	0.53
Coefficient of skewness	0.23	−0.06	−0.051
Coefficient of kurtosis	9.2E-05	4.0E-05	5.3E-05

dimensions of each feature map. After convolution and pooling the layers, the feature maps are flattened in the fully connected layer before a prediction is made using linear classifiers. The output of each convolution operation at each layer is activated using activation functions before applying pooling operations.

The FCNN model was trained with proximate analysis results and NIR imaging of the RDF. In the model with $28 \times 28 \times 1$ images as input, the output of the fully connected layer was used to predict the amount of moisture, ash, and volatile matter in the RDF samples. The visuals used in the train (63% of all data), validation (27% of all data), and test (10% of all data) phase are separated so that the images derived from each sample are used together.

Evaluation metrics

The root mean square error (RMSE), ratio of performance to deviation (RPD), R^2 , error variance (σ^2), and standard deviation values were calculated to measure the performance of the studies. The RMSE is the value expressing the error of estimation, and a low value is suitable to get an accurate model. R^2 is a parameter that expresses the proximity of the predicted value to the real value. It is the most important function in the model and is expected to be close to 100% for desired results. As another parameter, standard deviation expresses the spread of values. The Eqs. 10–14 give the RMSE, RPD, R^2 , and error variance equations. The results acquired from the models are given in the following tables.

$$RMSE : \text{root mean squared error} = \sqrt{\sum_{i=1}^n \frac{(y_{\text{observed},i} - y_{\text{predicted},i})^2}{n}} = \sqrt{MSE} \quad (10)$$

$$RPD : \text{ratio of performance to deviation} = SD/RMSE \quad (11)$$

$$SD : \text{standard deviation} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (y_{\text{observed},i} - y_{\text{mean}})^2} \quad (12)$$

$$R^2 : R\text{-squared} = 1 - \frac{\sum_i (y_{\text{observed},i} - y_{\text{predicted},i})^2}{\sum_i (y_{\text{observed},i} - y_{\text{mean}})^2} \quad (13)$$

$$\sigma^2 : \text{error variance} = \frac{\sum_{i=1}^n (x_{\text{error},i} - \bar{x}_{\text{mean}})^2}{n} \quad (14)$$

Results and discussion

The results obtained in the study are given under four main headings; proximate analysis results, statistical analysis results, measurement uncertainty results, and deep learning modeling results.

Proximate analysis results

The minimum and maximum values of the proximate analysis of the RDF samples were compared with the literature values (Table 2). According to the comparisons, the literature study results are within a variable range. The reason for this is entirely due to the different types of waste in the content of the RDF. When the results of this study are compared with the literature studies, it can be seen that the moisture and ash values in this study are in the extremely low range, and the volatile matter values are high as desired. The results of this study are similar to those of Miskolczi et al. (2011), Efika et al. (2015), and Hashem et al. (2019), who work with plastic and rubber-containing waste. Therefore, the values

Table 4 Test results for the FCNN and ResNet models

Parameter	Model	R^2	Error variance, %	Standard deviation, %	RMSE, %	RPD
Moisture	FCNN	0.979	0.0140	0.1183	0.0253	4.6758
	ResNet	0.929	0.0712	0.2664	0.2876	0.9262
Ash	FCNN	0.983	0.1037	0.3218	0.4897	0.6571
	ResNet	0.887	1.4566	1.2047	1.3258	0.9086
Volatile matter	FCNN	0.952	0.5430	0.7361	1.1191	0.6577
	ResNet	0.900	2.1106	1.4501	1.5650	0.9265

Table 5 Comparison of prediction results with the related literature

Reference	Fuel type	Number of samples	Moisture (%)			Ash (%)			Volatile matter (%)		
			R^2	RMSE	RPD	R^2	RMSE	RPD	R^2	RMSE	RPD
Xiao and Le 2020	Coal	158	0.960	1.67	na	0.950	3.58	na	0.970	3.54	na
Le et al. 2018	Coal	100	0.970	0.8318	na	0.959	1.7461	na	0.989	2.2743	na
Rambo et al 2020	Coffee	49	0.720	1.308	5.2	0.790	0.385	5.4	na	na	na
Posom and Sirisomboon 2015	Biomass	100	0.969	4.0	5.7	na	na	na	na	na	na
Posom and Sirisomboon 2017	Biomass	80	na	na	na	0.510	0.77	1.44	0.820	1.15	2.55
Kaihara et al. 2002	Coal	15	0.974	na	na	na	na	na	0.977	na	na
This study	RDF	55	0.979	0.0253	4.6758	0.983	0.4897	0.6571	0.952	1.1191	0.6578

na, not available

found as a result of the study are suitable for RDF to be burned in a cement plant.

Statistical analysis results

Histogram graphs of the results of the proximate analysis of 55 RDF samples for each parameter are shown in Fig. 11. These values are the average of three replicate analysis results. The graphs drawn for moisture and ash are asymmetric, and the graph drawn for volatile matter is a symmetrical histogram graph.

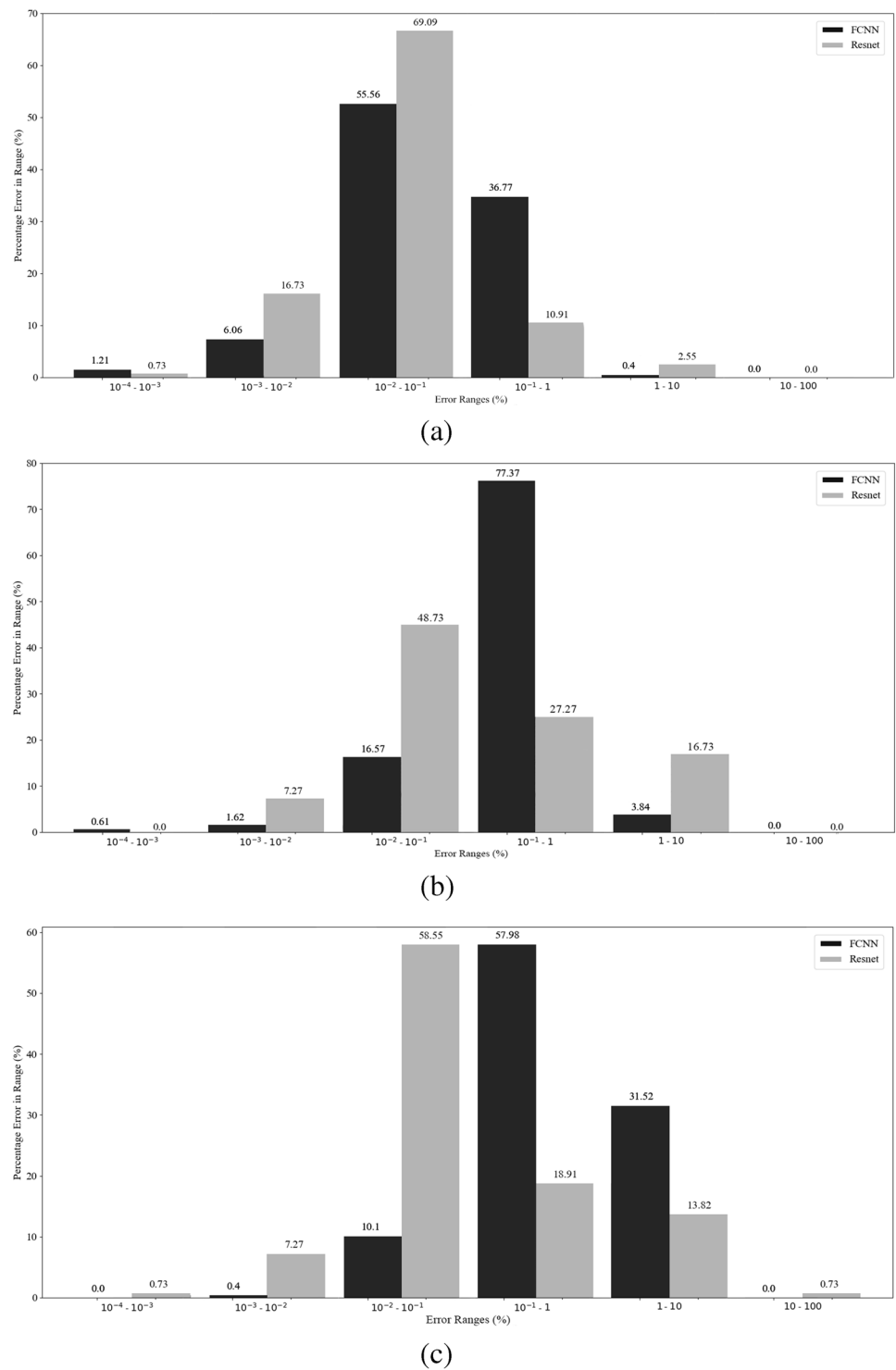
The results of the statistical calculations made using the data intervals and frequency values of histogram graphs are given in Table 3. The standard deviation (SD) value is proportional to the size of the distribution in a set of measurements. If the distribution is large, the SD value will be large. Therefore, the low SD values (0.09–0.43%) in Table 3 show the low distribution between the values of the three parameters. The coefficient of variation (CV) shows the relative distribution of different data sources. In light of this explanation, it can be concluded that the distribution of the values for the moisture is higher than those of ash and volatile matter, since the CV values of these parameters are 9.3, 2.64, and 0.53%, respectively. When a frequency distribution is asymmetrical, it is often described as a distorted frequency distribution. The coefficient of skewness (CS) shows the symmetry of a data stack based on its mean value; in fact, it is a symmetry. If the distribution is symmetrical, it means normal distribution, and the skewness is zero. On the other hand, if the skewness is lower than zero (negative skew) it means left skew, whereas for skewness higher than zero it means right skew (positive skew). According to the CS values in Table 3, it can be said that there is a normal distribution for the moisture values (0.23), whereas the ash (−0.06) and the volatile matter values (−0.051) show a left skew distribution. Another parameter that shows the shape of distribution is the coefficient of kurtosis (CK). The CK value of the normal distribution is zero, whereas

a CK value higher than zero shows a sharp peak, and a CK value lower than zero shows a flatter shape. In light of these explanation, since all the CK values in Table 3 are close to zero, it can be concluded that they all have a similar shape to normal distribution (Bellon-Maurel et al. 2010).

Deep learning modeling results

Within the scope of this study, two different studies were conducted based on visualization of spectrometer data with different methods, and then using them in moisture, ash, and volatile matter content estimation with two different deep learning models. The results of the study are given in Table 4. In addition, the distribution charts of the estimation deviations can be seen in Figs. 12 and 13. For the FCNN results, looking at the figures and tables, most of the 495 (10% of all data for test) estimates made have deviation values of less than 1%. When looking at the R^2 scores, it can be seen that success is reflected in high accuracy rates above 95%. It can be seen that there is a high error rate between 1 and 10 for volatile matter according to Fig. 12. For the ResNet results, as seen in Fig. 13, there is an accumulation close to 0 and below 1, except for outliers. When the R^2 scores are considered, despite a success of around 90%, it seems satisfactory, and the FCNN shows better performance (> 0.95) than those of the ResNet. Standard deviation is dispersion around the observation set's own mean, and is a measure of precision (Meyer 2012). Both the models have small SD values showing the precision of the predictions. Looking at the errors in the volatile matter values, it can be seen that the error rate between 1 and 10 has decreased. The RMSE is a dispersion around a true value, and is a measure of accuracy (Meyer 2012). According to Table 4, all the RMSE values are low depicting the accuracy of the models for all of the parameters. For the RPD evaluation, there is no statistical or utilitarian basis as to how these thresholds were determined. The researchers

Fig. 12 Error distribution graphs of the FCNN and ResNet. **(a)** Moisture values. **(b)** Ash values. **(c)** Volatile matter values



have tended to use RPD classes to designate their models as “excellent” for $RPD > 2$. According to this, the RPD value for moisture is excellent, while the others are below 2 (Mcbratney and Minansy 2013). On the other hand, Maurel et al. (2010) question the use of the RPD and point out that the normalization in the RPD only

works assuming normally distributed values. In light of these explanations, the RPD is meaningful for moisture prediction, since it shows a normal distribution, and it can be said that moisture prediction with the FCNN is successful with a high RPD value of 4.67. However, the RPD is not an appropriate metric to explain the success

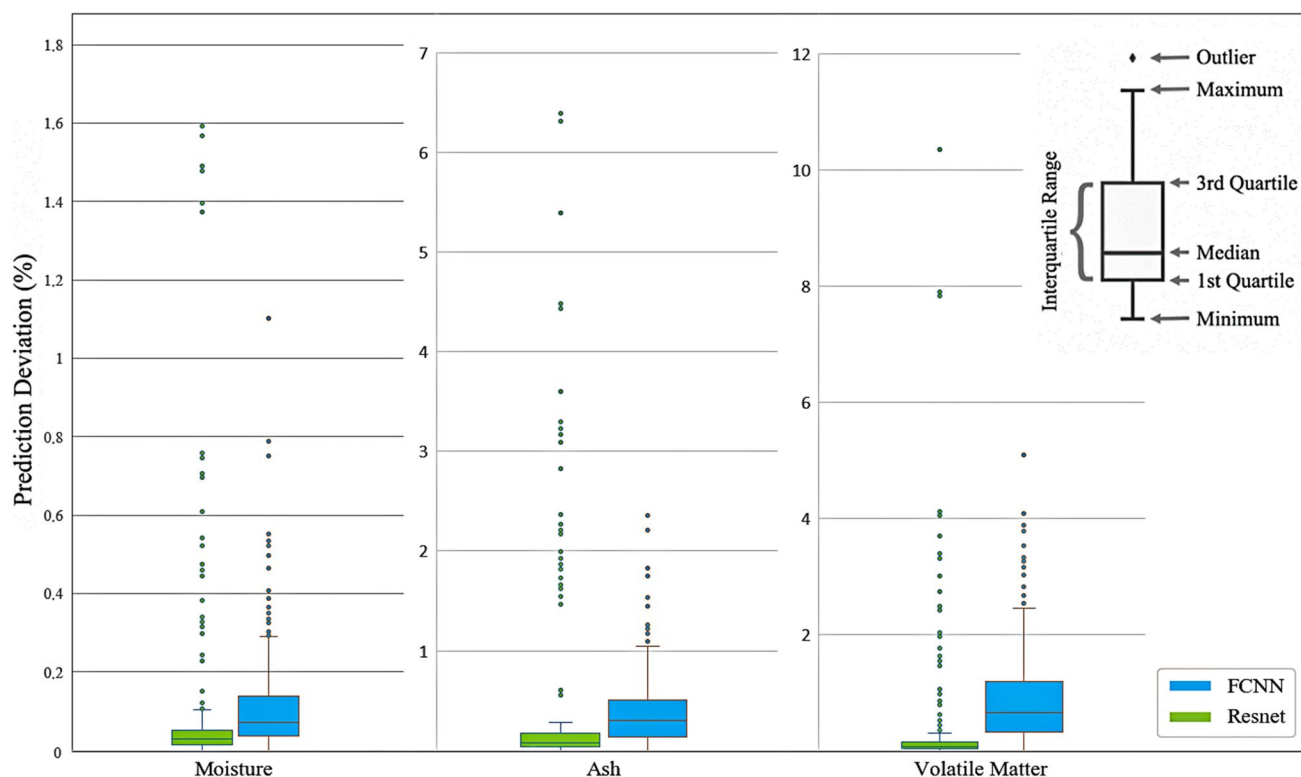


Fig. 13 Prediction deviation of the FCNN and ResNet

of the prediction of ash and volatile matter, since they do not show a normal distribution (please see Table 3).

The R^2 , RMSE, and RPD values obtained from the FCNN were compared with studies on coal and biomass, since there is no study related to waste in the literature, and these are given in Table 5. According to the table, as desired, while the R^2 values of the moisture and ash parameters are higher than those in the literature, the RMSE values are also quite low. Moisture in the material consisted of inherent moisture and free moisture. Free moisture is the moisture that is held on the surface, and inherent moisture is the moisture held within the pore structure and capillaries of the material. Both the inherent moisture and the free moisture can absorb an NIR wave, but no specific position was assigned to the inherent moisture in the NIRS of the material (Wang et al. 2014). From this point of view, it is thought that the prediction values are better than coal, since RDFs contain a large amount of free moisture. The ash content in the material is related to the presence of inorganic matter, which is not an NIR absorber, and the spectra from the NIR will not theoretically vary as the types and amounts of inorganic matter change. However, the NIR spectra may change due to the relationship between the inorganic matter and the organic matter in the RDF.

When the R^2 values in the volatile matter estimation results are examined, it can be seen that the values are lower than the studies on coal in the literature. Volatile content refers to the material which becomes mostly combustible gas upon combustion (after drying) and is generally equated with the organic (hydrocarbon-bearing) fraction. Therefore, it is in many ways comparable to the aliphatic content in terms of expected spectral correlations (Wang et al. 2014). Since RDF has a heterogeneous structure, the R values for the volatile matter were relatively low compared to coal. However, when the error values are examined, it can be seen that lower values are obtained than those in the literature, except ash for coffee. A number of the literature studies in Table 5 do not have SD or RMSE values, and it was not possible to make RPD calculations. It can also be seen that the RPD values given for moisture are close to each other. When the results are evaluated together, most of the acquired estimates have a deviation rate of less than 1%; the reason why the number of deviations in the volatile matter seems high is because the actual values of the sample are also large compared to another parameter. On the other hand, in the FCNN, where the estimation deviations are accumulated in smaller values, the standard deviation of errors is larger due to the presence of outliers at the same time.

Conclusion

DL models are used to classify various items or to estimate the amount of content. Nowadays, with the use of waste as fuel in various sectors, DL makes a great contribution to minimizing the problems that arise in the analysis that needs to be conducted immediately. Proximate values have an important place for calorific value estimation, which is one of the important parameters for waste. In this study, the applicability of DL is investigated in order to find a quick and effective method to determine the moisture, ash, and volatile matter content of RDF samples used in the cement industry. Two DL models, FCNN and ResNet, were tested, and it was seen that FCNN is more successful in representing the prediction. FCNN resulted in accurate (low RMSE values) and precise (low SD values) predictions for the proximate analyses. These results demonstrate the potential of using NIR-supported DL for quick determination if the RDF is appropriate for combustion in a cement kiln by minimizing the required analyzing time, laboratory equipment, and labor.

The success of the model depends on the variety and number of RDF samples. Therefore, this study could be extended to make a more robust model by inputting more samples with high variation. In addition to this, this study can be used as a baseline for an early warning system. As a conclusion, a successful and online prediction of the moisture, ash, and volatile matter content of RDF will provide significant benefits, in terms of both environmental and economic aspects.

Abbreviations CNN: Convolutional neural network; CK: Coefficient of kurtosis; CS: Coefficient of skewness; CV: Coefficient of variation; DL: Deep learning; FCNN: Fully convolutional neural network; FFT: Fast Fourier transform; ML: Machine learning; NIR: Near infrared spectroscopy; RDF: Refuse-derived fuel; SD: Standard deviation; STFT: Short time Fourier transform

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Declarations

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